

Logical operators:

1. && - Logical and
2. || - Logical or
3. ! – not

&&, || are used to combine two or more expressions into a single expression.

! used for negation i.e true becomes false and false become true.

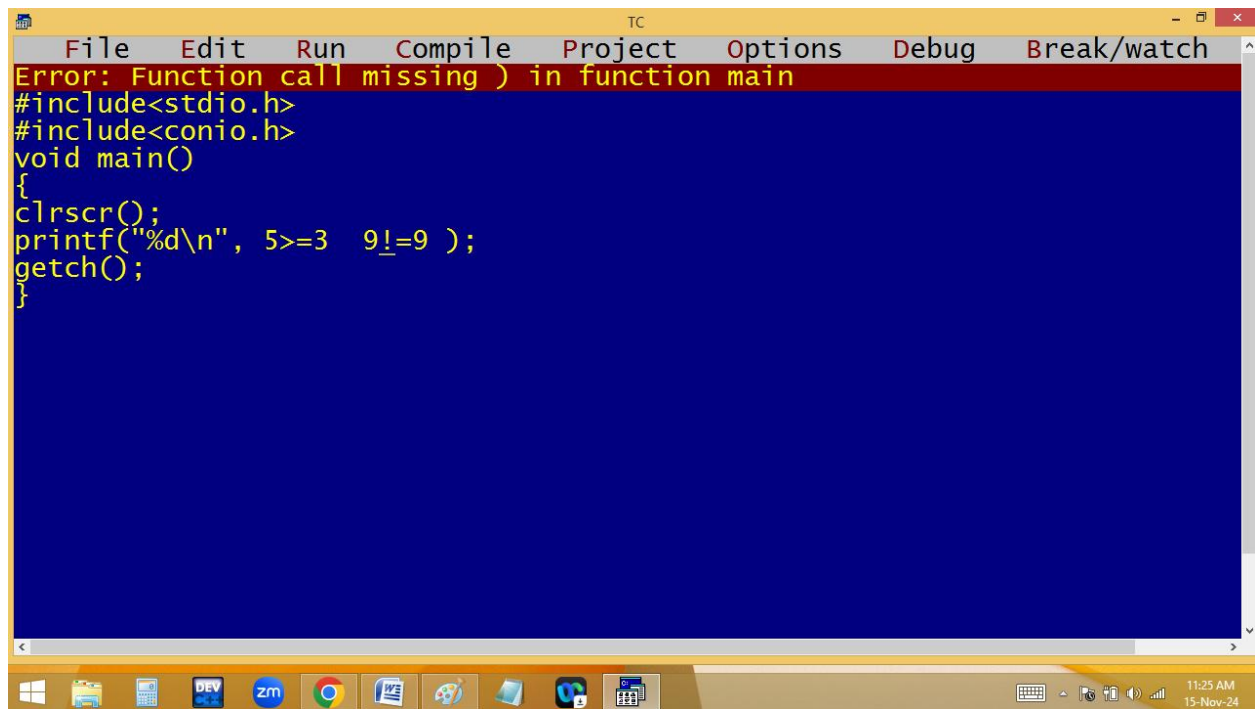
Note: In C other than 0 anything is 1 i.e. true

Truth tables:

Operator	Expression1	Expression2	Expression3
&& - and	1	1	1
	1	0	0
	0	1	0
	0	0	0
- or	1	1	1
	1	0	1
	0	1	1
	0	0	0

!true = false

!false = true



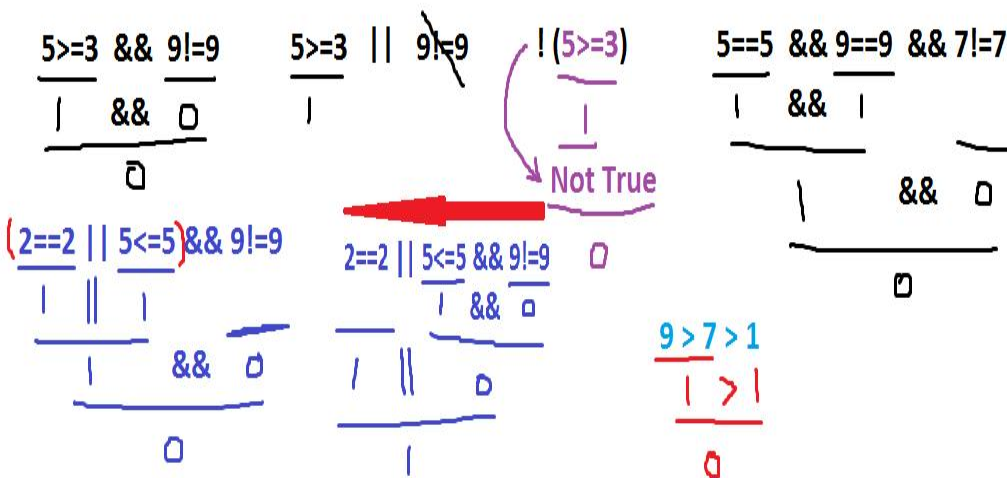
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A red error message banner at the top reads "Error: Function call missing) in function main". The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5>=3  9!=9 );
getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the date and time as "11:25 AM 15-Nov-24".

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 26 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5>=3 , 9!=9 );
printf("%d, %d\n", 5>=3 , 9!=9 );
printf("%d\n", 5>=3 && 9!=9 );
printf("%d\n", 5>=3 || 9!=9 );
printf("%d\n", !(5>=3) );
printf("%d\n", 5==5 && 9==9);
printf("%d\n", 5==5 && 9==9 && 7!=7 );
printf("%d\n", 2==2 || 5<=5 && 9!=9 );
printf("%d\n", (2==2 || 5<=5) && 9!=9 );
printf("%d\n", 9 > 7 > 1 );
printf("%d\n", 9 > 7 > 0 );
getch();
}
```

```
TC
1
1, 0
0
0
1
0
1
0
0
1
0
0
1
0
0
1
1
1
```



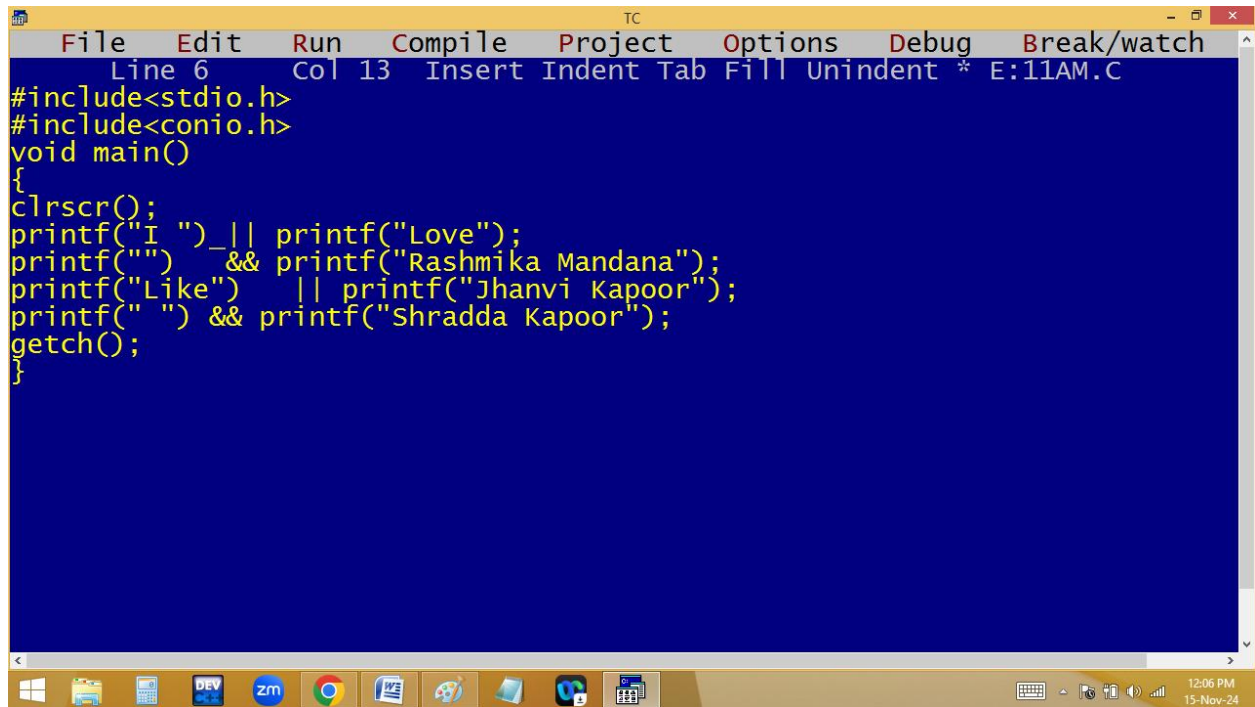
Note: In $||$ operation when left exp true, right exp not checked.
In $\&\&$ operation when left exp false, right exp not checked.

```

TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 5 && -5 );
printf("%d\n", 3 && 0 );
printf("%d\n", 1.1 || 2.2 );
printf("%d\n", 'a' && 'b' );
printf("%d\n", !'a' && 'b' );
printf("%d\n", 'a' || !'b' );
getch();
}

```

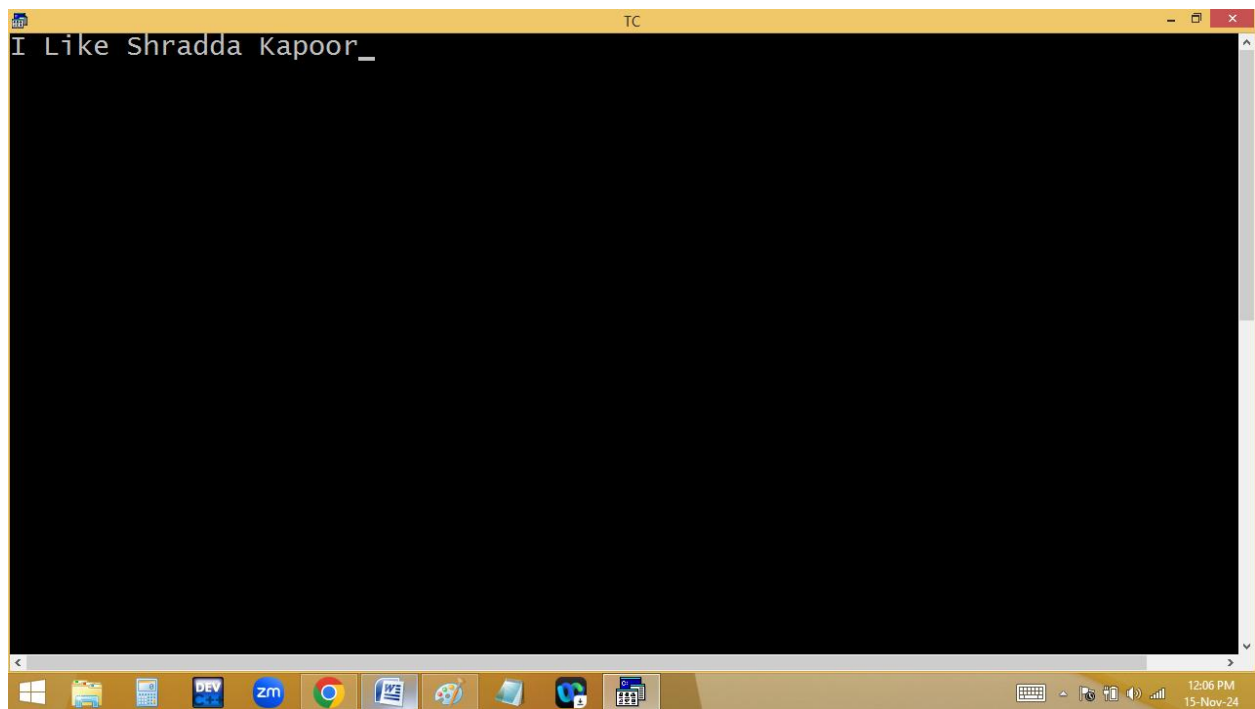
$$\begin{array}{r}
 \underline{5 \&\& -5} \\
 \underline{1 \&\& 1} \\
 1
 \end{array}
 \quad
 \begin{array}{r}
 \underline{3 \&\& 0} \\
 \underline{1 \&\& 0} \\
 0
 \end{array}
 \quad
 \begin{array}{r}
 \underline{1.1 || 2.2} \\
 1
 \end{array}
 \quad
 \begin{array}{r}
 \underline{'a' \&\& 'b'} \\
 \underline{97 \quad 98} \\
 \underline{1 \&\& 1} \\
 1
 \end{array}
 \quad
 \begin{array}{r}
 \underline{!'a' \&\& 'b'} \\
 \underline{97 \quad 98} \\
 \underline{!1 \&\& 1} \\
 \underline{0 \&\& 1} \\
 0
 \end{array}
 \quad
 \begin{array}{r}
 \underline{'a' || !'b'} \\
 \underline{97} \\
 1
 \end{array}$$



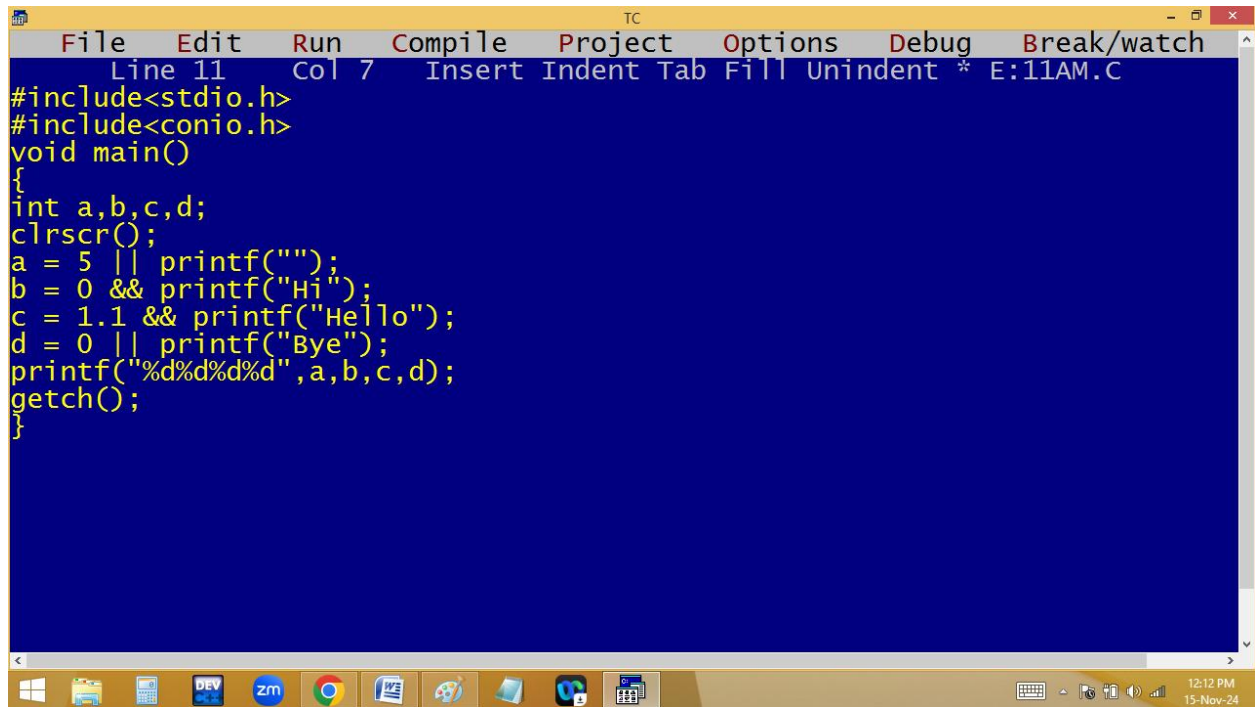
The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 6 Col 13 Insert Indent Tab Fill Unindent * E:11AM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("I ")_|| printf("Love");
printf("")_&& printf("Rashmika Mandana");
printf("Like")_|| printf("Jhanvi Kapoor");
printf(" ")_&& printf("Shradda Kapoor");
getch();
}
```

The Windows taskbar at the bottom shows icons for various applications including DEV, zm, Chrome, and the system clock displays 12:06 PM on 15-Nov-24.



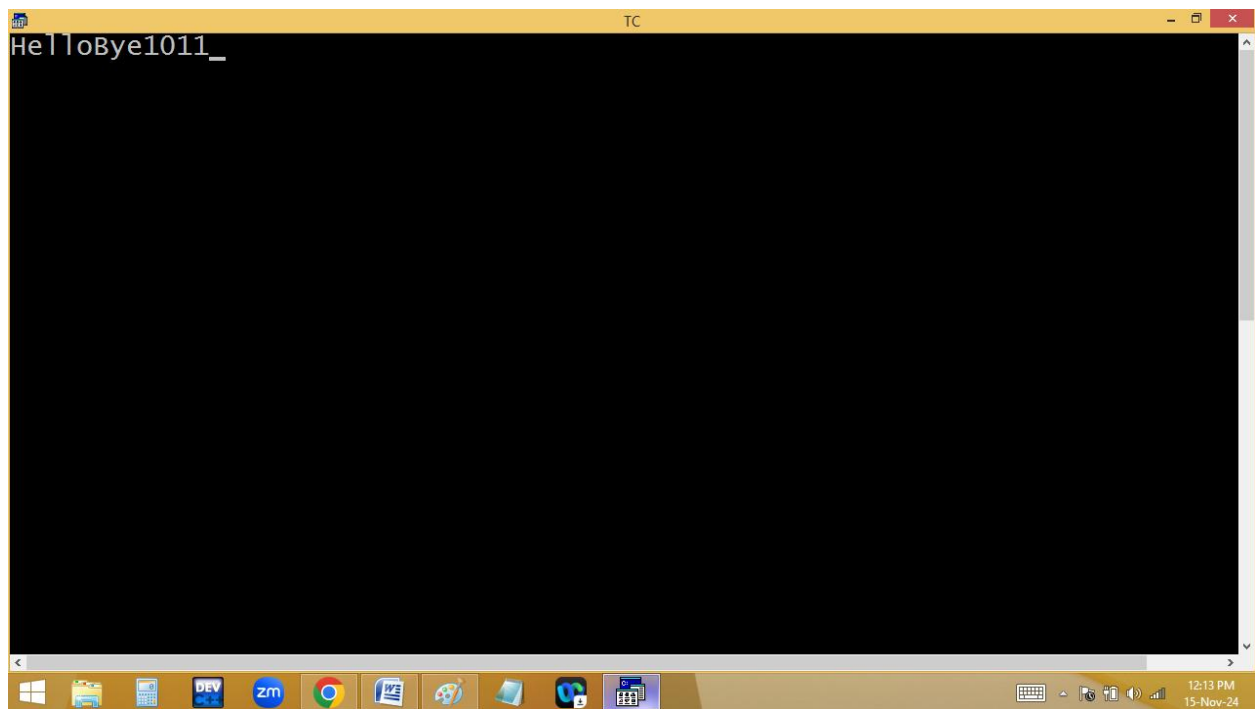
The screenshot shows the Turbo C++ (TC) IDE with a black background. The output of the program is displayed as 'I Like Shradda Kapoor_'. The Windows taskbar at the bottom is identical to the first screenshot, showing the same application icons and system clock.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar and a menu bar containing File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 11 Col 7' and 'Insert Indent Tab Fill Unindent * E:11AM.C'. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a = 5 || printf("");
b = 0 && printf("Hi");
c = 1.1 && printf("Hello");
d = 0 || printf("Bye");
printf("%d%d%d%d",a,b,c,d);
getch();
}
```

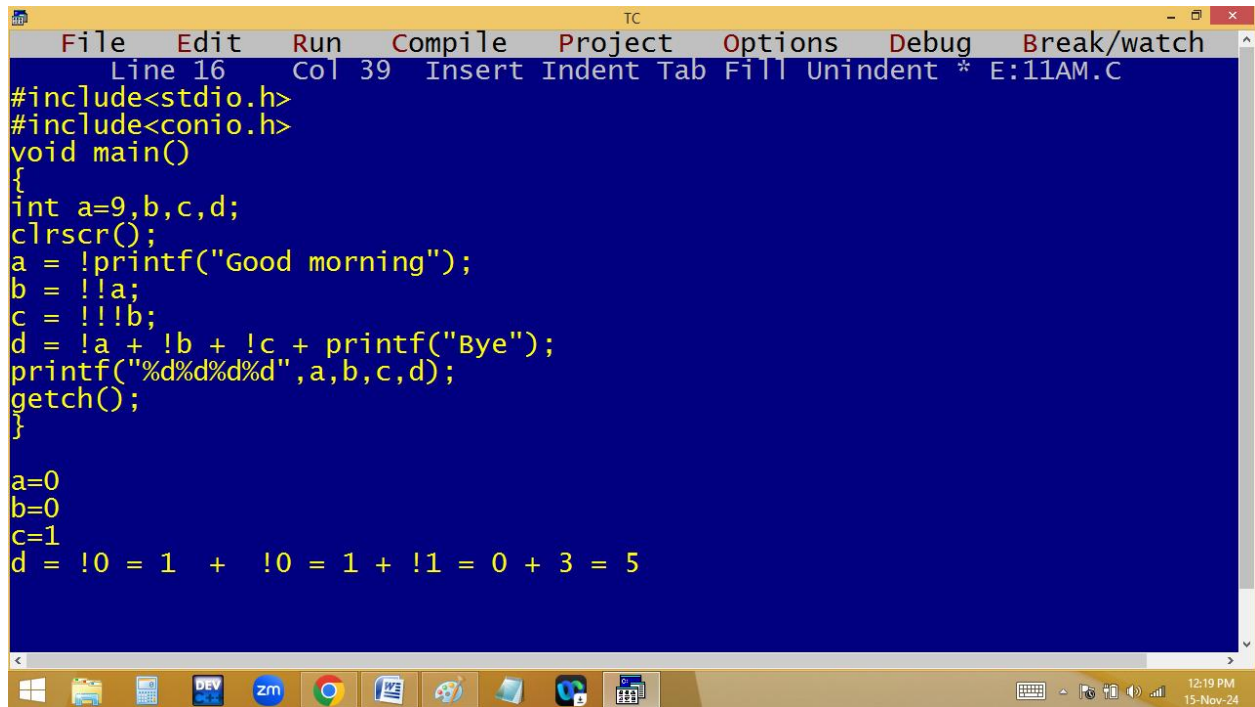
The Windows taskbar at the bottom shows various application icons and the system clock displaying 12:12 PM on 15-Nov-24.



The screenshot shows the Turbo C++ (TC) IDE with a yellow title bar. The main editing area has a black background and displays the output of the program: 'HelloBye1011_'. The Windows taskbar at the bottom shows various application icons and the system clock displaying 12:13 PM on 15-Nov-24.

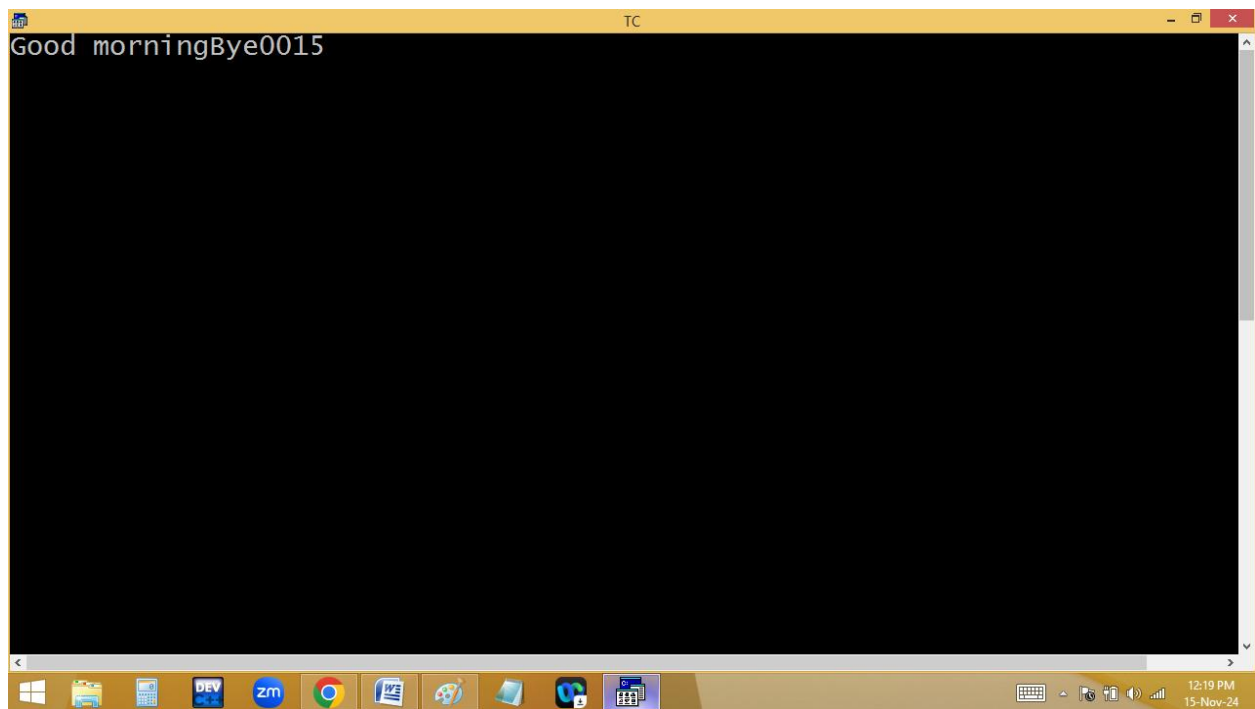
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 31 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9,b,c,d;
clrscr();
a = !printf("Good morning");
b = !!a;
c = !!!b;
d = !a + !b + !c + printf("Bye");
printf("%d%d%d%d",a,b,c,d);
getch();
}
```

```
TC
Good morningBye0015
12:15 PM
15-Nov-24
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 39 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9,b,c,d;
clrscr();
a = !printf("Good morning");
b = !!a;
c = !!!b;
d = !a + !b + !c + printf("Bye");
printf("%d%d%d%d",a,b,c,d);
getch();
}

a=0
b=0
c=1
d = !0 = 1 + !0 = 1 + !1 = 0 + 3 = 5
```



```
TC
Good morningBye0015
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int a=5, b=9, c=10;
clrscr();
c = a=1 && b=2;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
```

c = a=1 && b=2;

1 && 9
1 && 1

c = a = 1 = 2

constant=constant=error

scanf(): It is a predefined function available in stdio.h

It is used to read the values from user at runtime.

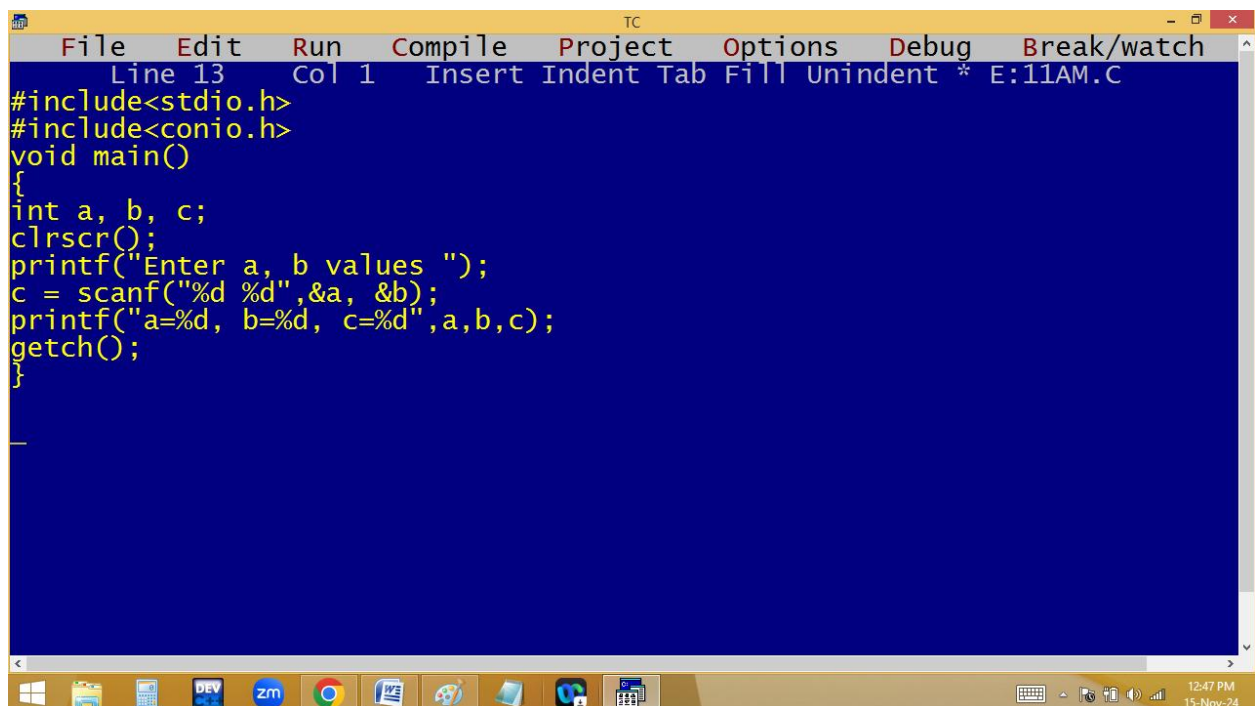
In scanf, f means formatted.

Scanf always read the inputs using standard input device i.e. keyboard.

Syntax:

int scanf("conversion characters / format specifiers",
&variable, &var,);

Write a c program to find no of conversion characters in scanf():

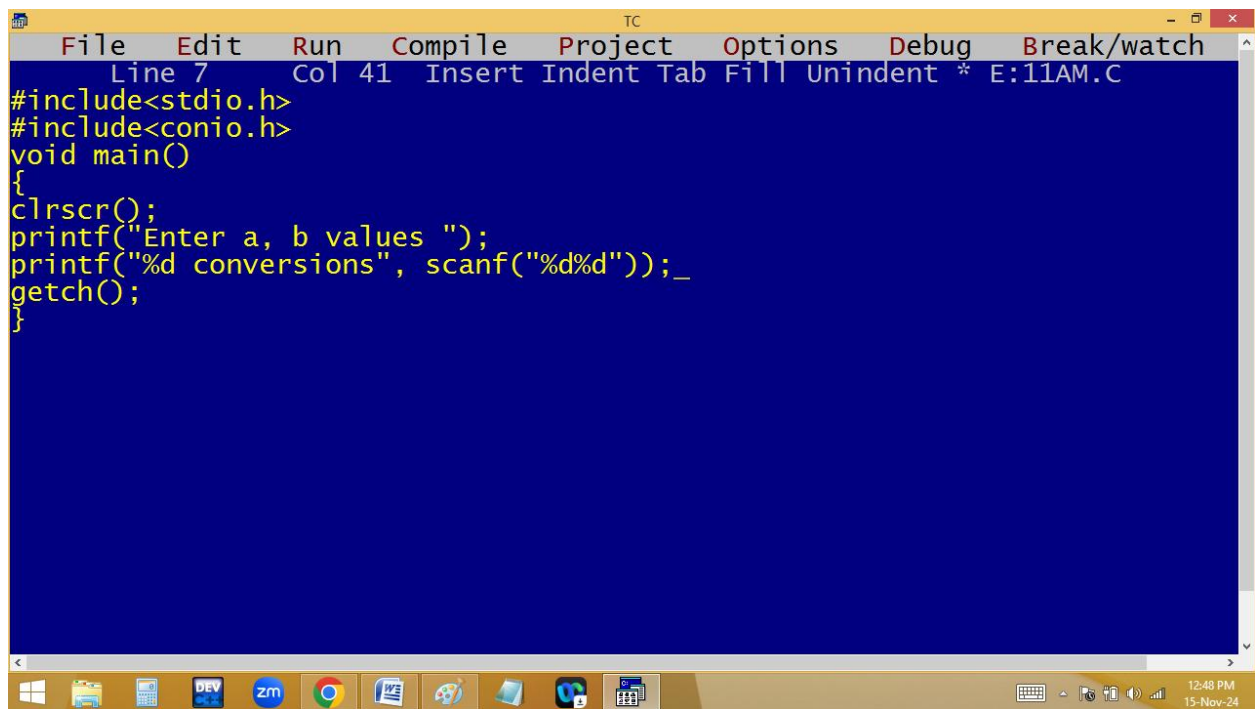


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b, c;
clrscr();
printf("Enter a, b values ");
c = scanf("%d %d",&a, &b);
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
```



A screenshot of a Turbo C++ (TC) console window. The window has a yellow title bar with the text "TC" and standard Windows window controls. The main area is black with white text. The text displayed is "Enter a, b values 9 5" on the first line and "a=9, b=5, c=2" on the second line. At the bottom, there is a Windows taskbar with various icons including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder icon. The system tray on the right shows the time as 12:47 PM and the date as 15-Nov-24.

```
Enter a, b values 9 5
a=9, b=5, c=2
```



A screenshot of a Turbo C++ (TC) editor window. The window has a yellow title bar with the text "TC" and standard Windows window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. Below the menu bar is a status bar that reads "Line 7 Col 41 Insert Indent Tab Fill Unindent * E:11AM.C". The main area has a blue background with yellow text. The code displayed is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Enter a, b values ");
printf("%d conversions", scanf("%d%d"));_
getch();
}
```

At the bottom, there is a Windows taskbar with various icons including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder icon. The system tray on the right shows the time as 12:48 PM and the date as 15-Nov-24.

```
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 41 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Enter a, b values ");
printf("%d conversions", scanf("%d%d"));_
getch();
}
```

